



DATE

23 September 2026

SOLAR SYSTEM AUDIT REPORT — MR STEPHEN UKAEGBU

CUSTOMER

Mr Stephen Ukaegbu

SYSTEM

6 kW Hybrid Inverter / 10 kWh Lithium Battery

PRIMARY FINDING

Undersized Solar Panel Array

CONNECTED LOADS

- 5 Ceiling Fans
- 2 Freezers
- 1 HP Inverter Air Conditioner

OPERATING HOURS

The system is used mainly during the daytime and evening hours. The connected appliances are not operated simultaneously, and the actual load varies depending on the appliances in use.

OBSERVATION

During system inspection, it was observed that the installed solar panel capacity is adequate for the 10 kWh lithium battery and 6 kW hybrid inverter under the current operating conditions.

However, an enhancement of the existing PV capacity will improve battery charging performance and further increase the uptime of the inverter and battery system for the customer.

RECOMMENDATION

The existing solar panel array should be enhanced by increasing the total PV capacity to increase the uptime of the inverter and battery system. This will improve charging performance and further extend the available operating hours of the system during normal solar hours.

CONCLUSION

The current PV capacity needs to be increased to meet the desired capacity of the customer for increased uptime hours. A PV upgrade by increasing the quantity of solar panels is recommended.